Hongzheng Tian

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Education

- **University of California, Irvine (UCI), CA, USA** Sep. 2022 - Jun. 2027 (expected)
  Program: Ph.D. in Computer Engineering
- **University of California, Irvine (UCI), CA, USA** Sep. 2020 - Jun. 2022
  Program: M.S. in Electrical and Computer Engineering
- **University of California, Riverside (UCR), CA, USA** Aug. 2019 - Jun. 2020
  Program: UCR-Graduate Preparation Program in Engineering (UCR-GPP-E)
- **University of British Columbia (UBC), BC, Canada** Jul. 2018 - Aug. 2018
  Program: Vancouver Summer Program in Communication Systems and Digital Systems Design with FPGA
- **Northwestern Polytechnical University (NWPU), Shaanxi, China** Sep. 2016 - Jun. 2020
  Program: Bachelor of Engineering's Degree in Microelectronics Science and Engineering

Scholarships & Awards

- **Best Paper Award - Artifact Track, ICPE'25** May. 2025
- **Nominated for Electrical Engineering and Computer Science Professor/Graduate Student/Undergraduate Student of the Year Award, UCI** Mar. 2024
- **Second-class Scholarship for Outstanding Student, NWPU** Dec. 2017 & Dec. 2018
- **Second Prize, The 19th Undergraduate Mathematical Contest in Modelling, NWPU** Jun. 2018

Skills

- **Languages:** C, C++, OpenMP, Python, Verilog HDL, VHDL and MATLAB
- **Software & Tools:** MLIR, LLVM, CUDA, MPI, HLS, Xilinx Design Tools, Synopsys, Cadence, Quartus, ModelSim, and MATLAB

Research & Publications

- **HeteroTask: Distributed Heterogeneous Task Runtime** Jan. 2025 - Apr. 2026

  Summary: Independently built a DAG execution framework with task-variant profiling, memory-aware device/variant selection, communication-cost modeling, and buffer lifetime planning. Implemented a C++/MPI runtime to execute planned task placements, dependency transfers, and buffer allocation/free operations. The project is complete; its scheduling concepts later informed CoVA.

- **HeteroBench: Multi-kernel Benchmarks for Heterogeneous Systems** Mar. 2024 - Aug. 2024

  Project jointly supervised by the CORSA Lab, UCI & Hewlett Packard Labs

  Advisor: Professor Sitao Huang & Dr. Alok Mishra

  Summary: Independently implemented a CPU/GPU/FPGA benchmark suite using Python, OpenMP, CUDA, and HLS. Built a JSON-configured Python driver with Jinja-generated Makefiles for backend-specific builds, benchmark execution, profiler integration, and experiment logging.

  Publication: ICPE'25 & SPEC RG Tools

  Award: Best Paper Award - Artifact Track

- **SAMT: Optimized Spatial Architecture Mapping Flow for Transformer Accelerators**

  Summary: A framework for optimizing dataflow mappings of Transformer inference workloads on spatial accelerators, combining mapping search with dynamic operator fusion to improve latency and energy efficiency across edge, mobile, and cloud platforms

  Publication: Haocheng Xu, Faraz Tahmasebi, Ye Qiao, **Hongzheng Tian**, Hyoukjun Kwon, and Sitao Huang. "Optimized Spatial Architecture Mapping Flow for Transformer Accelerators." arXiv:2410.07407, 2024.

- **CoVA: An MLIR-Based Compilation Flow for Versatile Architecture** Jun. 2023 - Present

  Doctoral research at UCI, continuing the project begun during the 2023 HPE internship

  Advisor: Professor Sitao Huang & Dr. Alok Mishra

  Summary: Built a numerical Python compiler integrating custom MLIR passes with upstream bufferization and lowering infrastructure for CPU LLVM/OpenMP and NVIDIA GPU LLVM/NVVM backends. Implemented loop and region storage analyses to prove independence and enable GPU-resident buffers where safe, GPU thread mapping, and fixed-group reduction trees with results independent of physical thread count for supported reductions. Implemented stream and temporary-buffer reuse and analysis-guided synchronization; evaluated correctness and performance with NPBench and paired measurements. FPGA support remains experimental.

  Publication: SC'24 ACM Student Research Competition Poster

- **A Sparsity-Aware Autonomous Path Planning Accelerator with Algorithm-Architecture Co-Design** Jan. 2023 - Sep. 2023

  Project jointly supervised by the CORSA Lab, UCI & the School of Cyberspace Science and Technology, Beijing Institute of Technology, BIT

  Advisor: Professor Sitao Huang & Professor Yanjun Zhang & Dr. Bo Yu

  Summary: FPGA-based acceleration of autonomous path planning, focusing on sparse matrix-vector multiplication and preconditioned conjugate gradient (PCG) computation for quadratic programming

  Publication: ACM TACO (2025), ICCAD'24, FPGA'24 Poster presentation

- **PyAIE: a Python-based Programming Framework for Versal ACAP AI Engines** Sep. 2022 - Jun. 2023

  Project at the CORSA Lab, UCI

  Advisor: Professor Sitao Huang

  Summary: Independently implemented a Python-based framework generating AI Engine kernel code, host code, and configuration scripts for AMD Versal ACAP.

  Publication: DAC'23 Late Breaking Results

- **Design Exploration of Associative Processor Implementation on FPGA** Apr. 2021 - Feb. 2022

  Project at the Lab of Center for Embedded and Cyber-physical Systems, UCI

  Advisor: Professor Fadi J. Kurdahi & Dr. Mohammed E. Fouda & Dr. Minjun Seo

  Summary: Built a new FPGA implementation of associative processor

  Publication: ISCAS'23 presentation & IEEE TCAS-II (2023)

Internships

- **CoVA: An MLIR-Based Compilation Flow for Versatile Architecture** Jun. 2023 - Sep. 2023

  Employer's name: Hewlett Packard Enterprise, **Job title:** Research Associate Intern

  Summary: Build an MLIR-based compiling flow for heterogeneous backend devices, enhancing code portability and efficiency

  Duties & Achievements:

  - Designed custom MLIR dialects to separate high-level program semantics from hardware-specific details
  - Implemented compiler analysis and transformation passes for a heterogeneous compilation flow

- **HeteroBench: A Benchmark Suite for Heterogeneous Computing Systems** Jun. 2024 - Sep. 2024

  Employer's name: Hewlett Packard Enterprise, **Job title:** Research Associate Intern

  Summary: Develop a benchmarking framework to evaluate performance and scalability across heterogeneous computing platforms

  Duties & Achievements:

  - Independently implemented CPU/GPU/FPGA benchmarks using Python, OpenMP, CUDA, and HLS
  - Implemented a JSON-configured Python driver with Jinja-generated Makefiles for backend-specific builds and benchmark execution
  - Integrated profiling commands, experiment parameters, and execution logs into the benchmark workflow

- **Design of MQ Decoder for a Real-Time JPEG2000 Player Based on FPGA** Sep. 2021 - Jan 2022

  Employer's name: IMBED LLC, **Job title:** Development Engineer trainee

  Summary: Designed a real-time JPEG2000 player for digital cinema applications

  Duties & Achievements:
  - Implemented the C code of the JPEG2000 decoder according to the JPEG2000 official document
  - Wrote and simulated the tier-1 entropy decoder (MQ decoder) in Verilog HDL
  - Learned from the relevant literature about the current improvement methods for MQ decoders
  - Optimized the basic version decoder to meet the timing and throughput constraints
  - Evaluated its performance, which is up to 450 MHz

Undergraduate Projects

- **Design of Digital Multi-Channel Lock-in Amplifier based on FPGA** Sep. 2019 - Mar. 2020

  Senior design at the Bourns College of Engineering, UCR

  Advisor: Professor Shane Cybart

  Summary: Co-designed a multi-channel lock-in amplifier based on FPGA which extract the desired signal with a known carrier wave from an environment that contains extremely noisy

  Duties & Achievements:
  - Designed the digital phase-locked loop to synchronize the external input reference signal
  - Designed the reference signal generator to generate the internal reference square signal
  - Designed the vector calculator to deal with the output vectors from the IIR filter
  - The final system can extract 0.5 uV tiny sinusoidal signal from 500 mVpp white noise

- **Design of Convolutional Neural Network (CNN) based on FPGA** Sep. 2019 - Dec. 2019

  Research at the Bourns College of Engineering, UCR

  Advisor: Professor Philip Brisk

  Summary: Assisted in the design of an FPGA-based Convolutional Neural Network for processing medical images

  Duties & Achievements:
  - Self-studied artificial intelligence and CNN related knowledge and got familiar with the C++ code of the CNN that the lab has completed
  - Learned how to improve the speed of CNN based on the FPGA
  - Learned how to use Vivado HLS to translate the high-level language to Verilog HDL and packaged the design as an IP for further tests and experiments

- **Design of a Q-Learning Algorithm to solve the Find-Path Problem** May. 2020 - Jun. 2020

  EE XRC260 Course project, UCR

  Advisor: Professor Sheldon Tan

  Summary: Built a program that can find the correct path of a maze by using the Q-Learning algorithm

  Duties & Achievements:
  - Studied the basic knowledge of Reinforcement Learning
  - Learned the basic algorithm of the Markov decision process and the Deep Q-Learning
  - Implemented the algorithm via MATLAB and designed a program to find the shortest and correct path out of the maze.

- **Design and Manufacture of AM/FM Radios** May. 2018 - Jun. 2018

  Electronic practice at Training Center of Engineering, NWPU

  Duties & Achievements:
  - Explored the principles and structures of AM/FM radios
  - Learned welding skills for electronic components
  - Welded a radio and completed related commissioning and assembly work

• **Undergraduate Mathematical Contest in Modelling, Northwestern Polytechnical University** May. 2018

- Design a mathematical model by using cluster analysis and hierarchical analysis to analyze the text from the topology structure and logical structure
- Use MATLAB programming to count the frequency of each character in each paragraph to determine whether the contents of a book were written by the same author

Activities

- **ICPE 2025: "Benchmarking in the Data Center: Expanding to the Cloud" Workshop.** Invited session chair and speaker; declined due to travel constraints.
- **SC24 ACM Student Research Competition.** Poster presentation, Atlanta, GA, 2024.
- **UCI GEECS Showcase.** Poster presentation, Irvine, CA, 2024.
- **FPGA 2024.** Poster presentation, Monterey, CA, 2024.
- **Young Fellow Program of the 60th Design Automation Conference.** Poster presentation, San Francisco, CA, 2023.
- **LATTE '23, Workshop on Languages, Tools, and Techniques for Accelerator Design.** Presentation, 2023. **Hongzheng Tian**, Shining Yang, Yoonha Cha, and Sitao Huang. "PyAIE: A Python-based Compiler Framework for Versal ACAP Platforms."

Related Courses

• At University of California, Irvine:

- EECS 222 Embedded System Modeling: A
- EECS 226 Embedded System Software: A
- EECS 221 Languages and Compilers for Hardware Accelerators: A

• At University of California, Riverside:

- EE XRC 133 Solid-State Electronics: A
- EE XRC 146 Computer Vision: A
- EE XRC 168 Introduction to Very Large Scale Integration (VLSI) Design: A+
- EE XRC 260 Seminar in Electrical Engineering (Advanced VLSI Design for Machine Learning and AI): A+

Teaching Assistantships

• **EECS 112L** Organization of Digital Computers Laboratory, Fall 2026.

• **EECS 112** Organization of Digital Computers, Fall 2023.

Extracurricular Activities

• **Theory Department Undersecretary & Photography Team Leader, Astronomical Society, NWPU** Aug. 2017 - Jun. 2018

- Taught starry sky photography knowledge and organized teammates to conduct starry sky photography during field inspections and contribute their works to various platforms of the university
- Taught and popularized astronomical knowledge by organizing related lectures and competitions
- Selected topics for the Starry Sky Carnival Competition while assuming the posts of question setter and judge for the Astronomical Joint Competition for Universities and Colleges in Shaanxi Province

• **Learning Department Secretary, Student Union of School of Software and Microelectronics, NWPU** Aug. 2017 - Jun. 2018

- Organized school activities and affairs with department members
- Solved learning problems for students to clarify their doubts
- Hosted learning experience exchange meetings and academic lectures for a good academic atmosphere
- Organized subject knowledge competitions and debate contests by selecting and training debaters
- Set questions for one advanced mathematics exam, invigilated it, and marked the exam papers